



Do Genres Feel Different? A Scene-Level Sentiment Analysis of Film Dialogue

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Abstract

There is growing interest in how different genres create distinct emotional experiences in relation to movie enjoyment and audience reviews [1, 2, 3]. Our project analyzed these differences by examining how dialogue sentiment changes across the course of a screenplay. We found that 1) genres differ in emotional experience, with especially clear differences in resolutions; 2) genre-specific terms often carry stronger emotional tone; and 3) genres differ in both emotional structure and emotional vocabulary.

Research Question

1. Do movie genres create distinct emotional arcs through dialogue?
2. Where in a film's story structure do genres differ most in sentiment?
3. Does genre-specific vocabulary carry stronger emotional tone than ordinary dialogue?

Dataset Description

1) **Film Corpus 2.0** - We used Film Corpus 2.0, which is a dataset of 960 scripts with dialogue separated from scene descriptions. We parsed each script into individual scenes using scene structure and extracted the spoken dialogue.

2) **Scene-Level Dialogue Dataset** - We converted the script data into a scene-level DataFrame. Each row corresponds to a scene. We used the relative scene position to compare films with different runtimes.

3) **TMDb Metadata** - We used the TMDb API to collect metadata for each film, such as genre and runtime. We matched and merged the scene DataFrame to its corresponding TMDb entry using title-matching based on title similarity and movie popularity.

Data Exploration

A. Unique Movies Per Genre

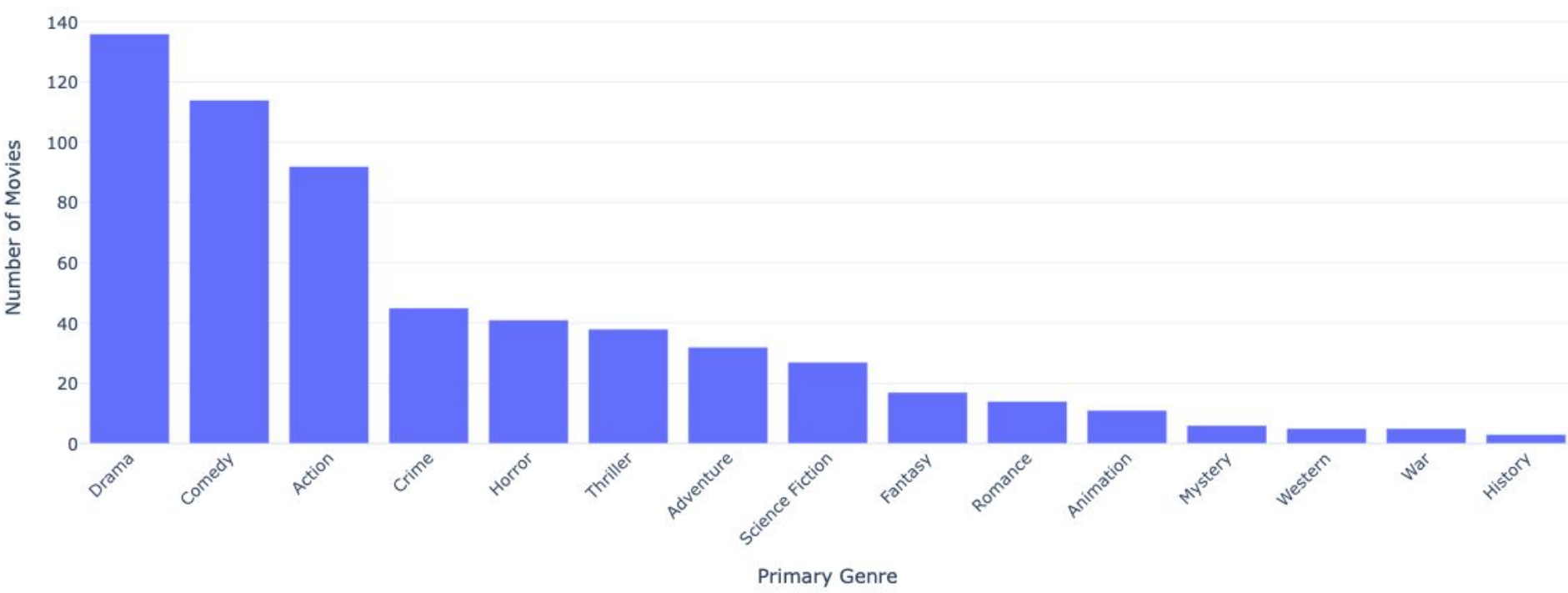
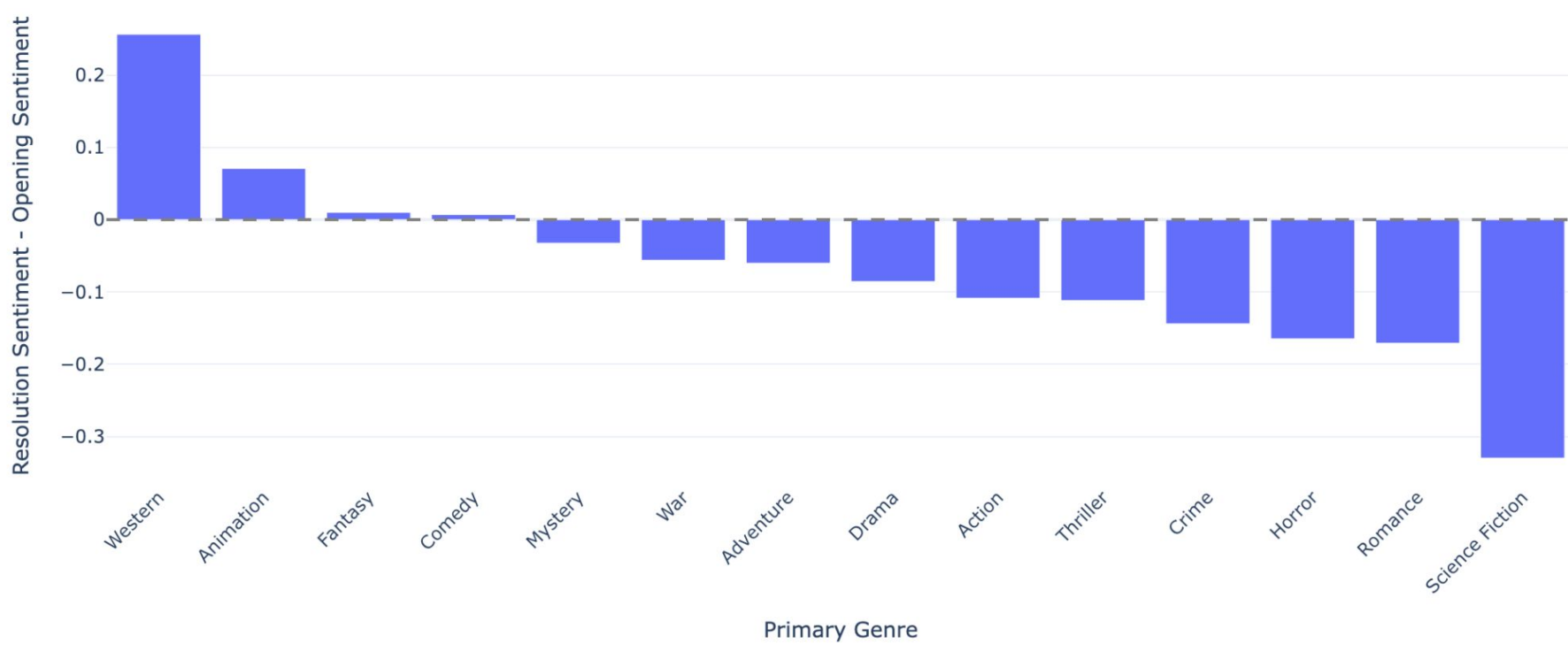


Figure 1. Exploratory analysis of film corpus. A) Drama, Comedy, and Action movies account for over half of the corpus.

B. Average Sentiment Change from Opening to Resolution



C. Emotional Arcs Across Movie Genres

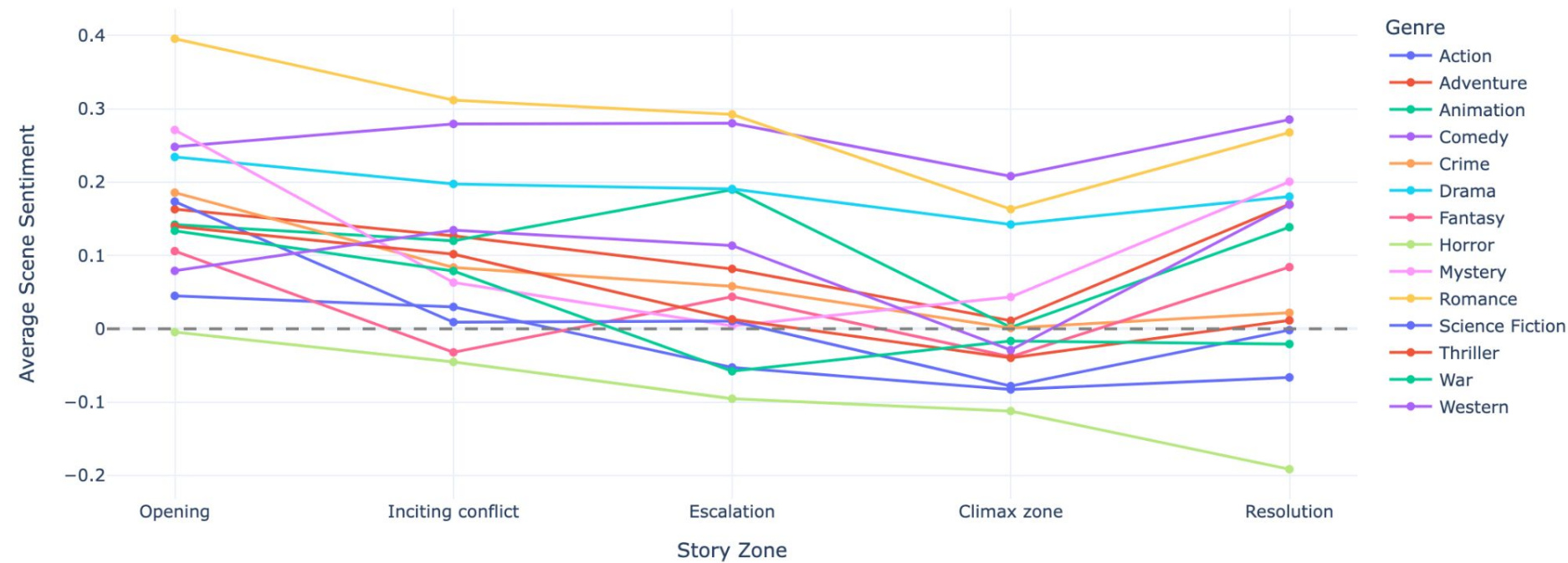


Figure 2. Sentiment Arc Analysis. We measured the average sentiment of each scene using VADER's Sentiment Intensity Analyzer (from -1 to 1) and used story position to compare emotional tone changes across genres. **B)** The change from opening to resolution shows how genres also differ in how they resolve emotional arcs. Western and Animation become more positive by the end, while Science Fiction, Romance, and Horror become more negative. **C)** Genre-level emotional arcs show that genres follow different emotional paths across the story. Romance, Comedy, and Drama stay relatively positive, while Horror remains the most negative and becomes even worse by the resolution.

Data Analysis

A. Original vs. TF-IDF-Filtered Sentiment ($r = 0.79$)



B. How TF-IDF Filtering Changes Measured Sentiment



C. Original vs. TF-IDF-Filtered Sentiment

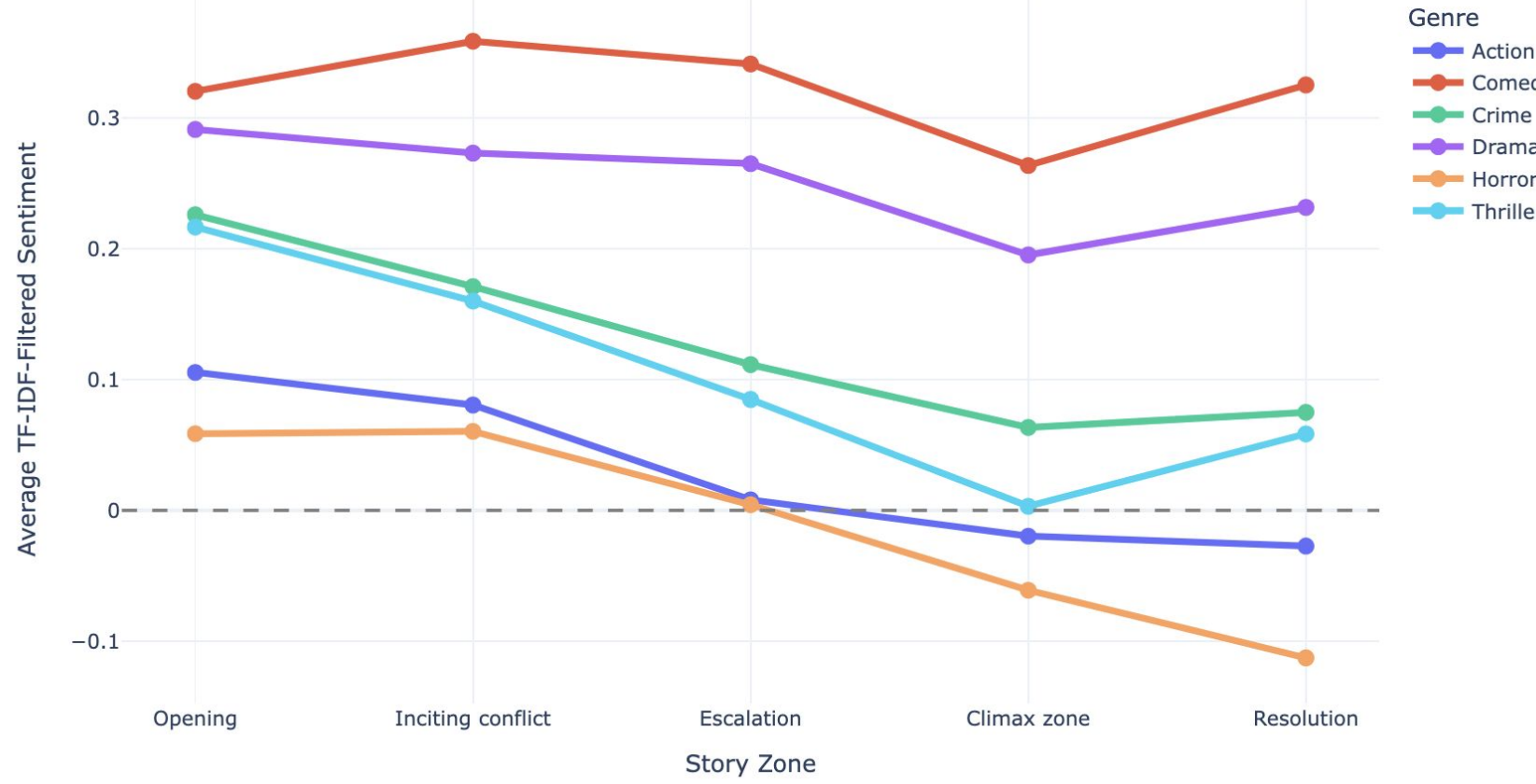


Figure 3. TF-IDF Analysis. We fit a TF-IDF vectorizer to all scene-level dialogue to isolate the most informative words for each genre. Each scene was filtered to keep only informative words, then sentiment was recalculated on the filtered text. **A)** Original and TF-IDF-filtered sentiment scores remained strongly correlated, showing that filtering preserved emotional signal. **B)** Across most story zones, filtered sentiment was higher than original sentiment. This indicates genre informative words carry stronger emotional tone. **C)** Genre-level emotional arcs remained visible after filtering. This suggests that genres differ in emotional vocabulary as well.

Conclusion

Our results show that movie genres create distinct emotional experiences through dialogue. Romance, Comedy, and Drama had the most positive sentiment, while Horror and Action were more negative. Genre-level emotional arcs also differed across story position, showing that emotion develops differently from opening to resolution. TF-IDF filtering preserved the main sentiment patterns, suggesting that genre-specific vocabulary carries meaningful emotional tone. These methods could help analyze whether a script matches genre expectations, while future work could connect emotional arcs to character dialogue, audience ratings, or box office success.

[1] Frangidis P, Georgiou K, Papadopoulos S. Sentiment Analysis on Movie Scripts and Reviews: Utilizing Sentiment Scores in Rating Prediction. Artificial Intelligence Applications and Innovations. 2020 May 6;583:430–8.
[2] Bartsch, A. (2012). Emotional Gratification in Entertainment Experience. Why Viewers of Movies and Television Series Find it Rewarding to Experience Emotions. Media Psychology, 15(3), 267–302.
[3] Mokryn, O., Bodoff, D., Bader, N. et al. Sharing emotions: determining films' evoked emotional experience from their online reviews. Inf Retrieval J 23, 475–501 (2020).